Positive Square Energy of Every Heptacyclic Cactus

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Abstract

Let G be a connected heptacyclic cactus of order n. We prove $s^+(G) > n$. Sharp block-additive doubly nonnegative optimization reduces the proof to the cycle multisets T^6Q and T^5PP, where $T = C_3$, $P = C_5$, and Q is arbitrary. A shared-triangle opening recurrence gives the quantitative margins needed for both residuals. If the shared-cut graph is disconnected, T^6Q is settled by an all-triangle leaf cluster. For T^5PP, an exact colored-partition audit has 46 proper rows: 41 are direct and five are closed structurally, including an entry-sensitive degree dichotomy for $T^5P \mid P$. If all cycles form one shared-cut cluster, T^6Q is settled by opening a leaf Q or splitting an internal Q, while T^5PP is settled by opening two pentagonal leaves or splitting an internal pentagon. Independent exact censuses reproduce the corrected totals $216, 224, 226, 227, 227$ for T^6Q and $560 = 557 + 3$ for T^5PP. Every construction is a vertex partition into induced subgraphs and permits arbitrary bridge connectors and arbitrary trees attached at arbitrary vertices.

1 Preliminaries and dependencies

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are bridges or cycles. Its cyclomatic number is the number of cyclic blocks. If the adjacency eigenvalues of G are $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \qquad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \qquad \sigma(G) = s^+(G) - |V(G)|.$$

For every vertex partition $V(G) = V_1 \,\dot\cup\, \cdots \,\dot\cup\, V_k$, induced-subgraph superadditivity gives

$$s^+(G) \ge \sum_{i=1}^{k} s^+(G[V_i]), \qquad \sigma(G) \ge \sum_{i=1}^{k} \sigma(G[V_i]). \tag{1}$$

This is the theorem of Akbari, Kumar, Mohar, and Pragada [1].

Write $T = C_3$, $P = C_5$, and

$$\delta = \sec(\pi/5) - 1 = \sqrt{5} - 2 < \frac{1}{4}, \qquad \delta_q = \sec(\pi/q) - 1 < 1 \quad (q \equiv 1 \pmod 4).$$

We use the following established packet bounds:

$$\sigma(T) > 0, \qquad \sigma(P) \ge -\delta, \qquad \sigma(C_q) \ge -\delta_q, \tag{2}$$

$$\sigma(TT) > 1, \qquad \sigma(TP) > 1 - \delta, \qquad \sigma(TC_q) > 1 - \delta_q. \tag{3}$$

For the lower-rank classes,

$$\sigma(H) \geq 0 \qquad \text{for every bicyclic or tricyclic cactus } H, \tag{4}$$

$$\sigma(H) > 0 \qquad \text{for every cactus of rank four, five, or six.} \tag{5}$$

For one shared-cut cluster we also use

$$\sigma(PP) \geq 1 - \frac{4}{3\sqrt{13}} > 0, \tag{6}$$

$$\sigma(TPP) > 6 - 2\sqrt{5} > \frac{3}{2}. \tag{7}$$

Here a symbol records the cyclic blocks retained by the packet, not its tree attachments. Every estimate allows arbitrary finite trees attached at arbitrary vertices; TP and TC_q also allow an arbitrary bridge connector. The one-, two-, and three-cycle bounds are proved in the bicyclic and tricyclic companion manuscripts [3, 4]; the lower-rank statements are the tetracyclic through hexacyclic cactus theorems [5, 6, 7]. Bound (6) is the exact shared-PP estimate in [3], and (7) is the shared C_3, C_5, C_5 phase estimate in [4].

We also need the shared all-triangle estimates

$$\sigma(A_1) > 0, \qquad \sigma(A_2) > 1, \qquad \sigma(A_3) > 2, \qquad \sigma(A_4) > 3, \tag{8}$$

where the triangles of A_r form one shared-cut cluster. For $r \leq 3$ these follow from the favorable-cycle Sachs theorem [9]. For four triangles the same theorem applies except to the central-triangle/three-petal incidence; the exact grouped-Sachs matching injection in [9, Proposition 4.11] gives the same strict bound there. Thus no cycle-packing hypothesis is hidden in (8).

2 Territories, intervals, and openings

Join two cyclic blocks in the *shared-cut graph* when they share a cut vertex, and call its connected components shared-cut clusters. In the block-cut tree contract the incidence subtree of each cluster to one marked node. Every cluster, including a singleton cyclic block, is marked. Take the minimal subtree spanning the marked nodes and suppress only unmarked degree-two nodes. The result is the *reduced cluster tree.*

Lemma 2.1 (Arbitrary connector territories)**.** *Let $S_1, \ldots, S_k$ be pairwise vertex-disjoint connected subtrees of the reduced cluster tree, and suppose every marked node belongs to exactly one S_i. Then $V(G)$ has a partition into connected induced territories $G_1, \ldots, G_k$ such that G_i retains exactly the cycles assigned to S_i. Cuts occur on actual bridges. Every connector remnant, Steiner branch, entry vertex, and attached tree has exactly one owner.*

Proof. Assign every remaining reduced-tree vertex to a nearest labelled subtree, breaking ties by a fixed order. Each label class in a tree is connected. Every edge between classes expands in the block-cut tree to a path consisting only of bridge blocks; cut one actual bridge on that path and assign its two remnants to the adjacent owners. At a Steiner branch assign the branch vertex and one initial segment to one owner and cut an actual bridge on every other incident segment. A component outside the cyclic hull has a unique attachment to it, since two attachments would create another cycle in the block-cut tree; assign it whole to the owner of that attachment. The resulting vertex sets are disjoint, connected, and induced, and retain exactly the asserted cycles. □

For one shared-cut cluster, its *cycle–cut incidence tree* is the bipartite tree whose nodes are cyclic blocks and shared cyclic cuts. A cycle node is adjacent to the distinct cyclic cuts on that cycle.

Lemma 2.2 (Intervals and private openings)**.** *Let C be a cycle node of a cycle–cut incidence tree.*

1. *If C has $d \geq 2$ distinct marked vertices, its vertices can be partitioned into d nonempty proper consecutive intervals, one owning each mark. Adjoining to an interval the corresponding component of $I - C$ gives a connected induced territory. The d territories partition all vertices assigned at C, every mark has one owner, and C is retained by none.*

2. *If v is private with respect to cyclic blocks, put v and every off-cycle tree branch rooted there into F. Then F is a nonempty tree and $\sigma(F) = -1$. The path $C - v$ retains every cyclic cut of C and belongs to the complementary induced territory.*

Both constructions remain valid with arbitrary hanging trees. An external connector entering at a private cycle vertex may be assigned to either adjacent interval; an entry at a marked cut or through an incidence branch follows the unique interval owning that branch.

Proof. For the first construction, list the distinct marks cyclically and cut one cycle edge in each selected gap between successive marks. Assign unmarked vertices in a gap to either adjacent interval. Each interval is a proper path, and each incidence branch follows its unique mark. For the second, $C - v$ is a path. The vertex v and all tree branches rooted at it form a nonempty tree. For every nonempty tree F, bipartite spectral symmetry gives $s^+(F) = |E(F)| = |V(F)| - 1$, hence $\sigma(F) = -1$. Unique attachment of hanging trees proves all ownership assertions. Crossing edges are discarded only by (1); no edge monotonicity is used. □

3 The shared-triangle recurrence

Lemma 3.1 (Shared-triangle recurrence)**.** *Let A_r be a connected cactus whose r triangular cyclic blocks form one shared-cut cluster, with arbitrary trees attached. Then*

$$\sigma(A_r) > b_r, \qquad \begin{array}{c|ccccccc} r & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline b_r & 0 & 1 & 2 & 3 & 2 & 1 & 0 \end{array}. \tag{9}$$

Proof. The values through four are (8). For $r \geq 5$, the cycle–cut incidence tree has a leaf triangle. It has one shared cyclic cut and two vertices private with respect to cyclic blocks. Open either private vertex by Lemma 2.2. The opened territory is a nonempty tree of surplus -1. Deleting the leaf cycle node, and suppressing its incident cut if it becomes irrelevant, leaves the other $r - 1$ triangle nodes connected in one shared-cut incidence tree. The complementary induced territory is therefore an A_{r-1} packet, with the surviving path fragment and all other trees merely attached to it. Consequently

$$\sigma(A_r) \geq \sigma(A_{r-1}) - 1.$$

Starting from the strict bound $\sigma(A_4) > 3$ gives successively $\sigma(A_5) > 2$, $\sigma(A_6) > 1$, and $\sigma(A_7) > 0$. □

The strict quantitative values $b_5 = 2$ and $b_6 = 1$ are essential below. Qualitative lower-rank positivity is never used to pay a tree-opening cost or a hostile odd-cycle deficit.

4 The exact DNN residual classification

For a connected edge-containing graph define

$$\kappa(G) = \sup_{\substack{M\succeq 0,\ M\geq 0\\ \mathbf{1}^T M\mathbf{1}>0}} \frac{4\left(\sum_{uv\in E(G)}\sqrt{M_{uv}}\right)^2}{\mathbf{1}^T M\mathbf{1}}.$$

The sharp cactus DNN theorem [8, 10] gives, for bridge-block count b and cycle lengths $\ell_1,\dots,\ell_r$,

$$\kappa(G) = b + \sum_j \kappa(C_{\ell_j}), \qquad \kappa(C_\ell) = \begin{cases}\ell, & \ell \text{ even},\\ \ell\sec^2(\pi/(2\ell)), & \ell \text{ odd},\end{cases} \qquad s^-(G)\leq\kappa(G). \tag{10}$$

For completeness, fold each nonedge entry of a DNN matrix into its two diagonal entries by adding $M_{uv}(e_u-e_v)(e_u-e_v)^T$. Rotational averaging on a cycle and its least Fourier eigenvalue give the cycle constants. Orthogonal gluing of folded Gram systems at a one-vertex sum, followed by Cauchy–Schwarz, proves block additivity. Finally, writing $A = A_+ - B$, the Schur product theorem makes $B\circ B$ DNN and gives

$$s^-/2 = -\sum_{uv\in E(G)} B_{uv} \leq \sum_{uv\in E(G)} |B_{uv}|, \qquad \mathbf{1}^T(B\circ B)\mathbf{1} = s^-;$$

the definition of κ yields $(s^-)^2 \leq \kappa(G)s^-$.

Put

$$\epsilon_\ell = \begin{cases}0, & \ell \text{ even},\\ \ell\tan^2(\pi/(2\ell)), & \ell \text{ odd}.\end{cases}$$

The odd sequence is decreasing, since after putting $u = \pi/(2x)$ the relevant derivative has the sign of $2u - \sin u\cos u > 0$. Moreover

$$\epsilon_3 = 1, \qquad \epsilon_5 = 5-2\sqrt{5} =: a, \qquad 3a<2, \qquad 2a>1, \qquad \epsilon_5+\epsilon_7<1. \tag{11}$$

These comparisons are exact. The middle two reduce, after squaring positive sides, to $169 < 180$ and $80 < 81$. Also $\cos(\pi/7) > 1-\frac{1}{2}(22/49)^2 > 7/8$, whence $\epsilon_7 < 7/15 < 2\sqrt{5}-4 = 1-\epsilon_5$; the last rational-radical comparison squares to $4489 < 4500$.

For a heptacyclic cactus, block counting and (10) give

$$b + \sum_{i=1}^{7}\ell_i = n+6, \qquad \sigma(G) \geq 6 - \sum_{i=1}^{7}\epsilon_{\ell_i}. \tag{4.1}$$

Indeed, $s^+ + s^- = 2|E(G)| = 2n+12$ and $s^- \leq n+6+\sum_i \epsilon_{\ell_i}$. If at most four cycles are triangles, the epsilon sum is at most $4+3a<6$. With exactly five triangles, an even remaining length contributes zero; if both remaining lengths are odd, every pair except $5,5$ contributes at most $\epsilon_5+\epsilon_7<1$. The pair $5,5$ survives because $2a>1$. With at least six triangles the multiset has the form $\{3,3,3,3,3,3,q\}$, $q\geq 3$. Thus (4.1) is strict outside exactly

$$T^6Q = \{3,3,3,3,3,3,q\} \quad (q\geq 3), \qquad T^5PP = \{3,3,3,3,3,5,5\}. \tag{12}$$

5 Disconnected shared-cut graph: the T^6Q family

Proposition 5.1. *If the cycles are T^6Q and the shared-cut graph is disconnected, then $s^+(G) > |V(G)|$.*

Proof. The reduced cluster tree is nontrivial and therefore has at least two marked leaves. At most one leaf cluster contains the designated block Q; this remains true when $Q = T$, because that block remains designated. Choose a Q-free leaf cluster A. It consists of r triangles in one shared-cut cluster, for some $1 \le r \le 6$. Cut an actual bridge toward the rest and use Lemma 2.1. The two connected induced territories are an A_r packet and a connected $(7-r)$-cyclic cactus containing Q. The full ledger is

r	triangle leaf	complementary territory	total
1	> 0	hexacyclic > 0	> 0
2	> 1	pentacyclic > 0	> 0
3	> 2	tetracyclic > 0	> 0
4	> 3	tricyclic ≥ 0	> 0
5	> 2	bicyclic ≥ 0	> 0
6	> 1	unicyclic Q	> 0

In the last row a nonhostile Q is nonnegative. If $q \equiv 1 \pmod 4$, the unicyclic territory, including all of its connector remnants and attached trees, has surplus at least $-\delta_q$; hence the total is $> 1-\delta_q > 0$. This proves every reduced-tree topology and every connector entry without an incidence census. □

6 Disconnected shared-cut graph: all 46 T^5PP partitions

Encode a shared-cut cluster by (t, p), its numbers of triangles and pentagons. Recursively choose the first nonzero pair, subtract it from $(5, 2)$, and require successive pairs to be lexicographically nondecreasing. This gives exactly 47 colored set partitions, including the one-cluster partition, and hence 46 proper partitions. Assign to each cluster the conservative exact-rational ledger

packet	T	P	TT	TP	PP	TTT	TPP	T^4	T^5	T^6
bound	> 0	$> -1/2$	> 1	$> 1/2$	> 0	> 2	$> 3/2$	> 3	> 2	> 1

and use ≥ 0 for every other rank-two or rank-three cluster and > 0 for every other cluster of rank four through six. These are consequences of (2)–(7), (5), and Lemma 3.1. A row is direct when the rational sum is positive, or is zero with a strict summand.

The resulting 41 direct rows, in canonical order, are

```
P|P|T|T|T|TT    P|P|T|T|TTT    P|P|T|TT|TT
P|P|T|TTTT    P|P|TT|TTT    P|P|TTTTT
P|T|T|T|T|TP    P|T|T|TP|TT    P|T|TP|TTT
P|T|TT|TTP    P|TP|TT|TT    P|TP|TTTT
P|TT|TTTP    P|TTP|TTT
PP|T|T|T|T|T    PP|T|T|T|TT    PP|T|T|TTT
PP|T|TT|TT    PP|T|TTTT    PP|TT|TTT    PP|TTTTT
T|T|T|T|TPP    T|T|T|TP|TP    T|T|T|TTPP
T|T|TP|TTP    T|T|TPP|TT    T|T|TTTPP
T|TP|TP|TT    T|TP|TTTP    T|TPP|TTT
T|TT|TTPP    T|TTP|TTP    T|TTTTPP
TP|TP|TTT    TP|TT|TTP    TP|TTTTP
TPP|TT|TT    TPP|TTTT    TT|TTTPP
TTP|TTTP    TTPP|TTT
```

Whole cluster territories and arbitrary connectors are supplied by Lemma 2.1. Exact `Fraction` enumeration leaves the following five rows, and no others:

$$\begin{gathered} P|P|T|T|T|T|T, \qquad TTP|P|T|T|T, \qquad TTTP|P|T|T, \\ TTTTP|P|T, \qquad TTTTTP|P. \end{gathered} \tag{13}$$

Lemma 6.1 (The four singleton-triangle rows)**.** *The first four rows of* (13) *have positive surplus for every reduced-tree topology and every connector entry.*

Proof. Write these rows as

$$A_k \mid T \mid \cdots \mid T \mid P_1,$$

where $A_0 = P_0$, while for $k = 2, 3, 4$ the packet $A_k = T^k P_0$ is the unique nontrivial pentagon-containing cluster. There is at least one singleton triangle. If one is a leaf of the reduced cluster tree, cut the first actual bridge toward the rest. The two induced territories are a strict triangular unicyclic cactus and a strict connected hexacyclic cactus.

Suppose no singleton triangle is a leaf. Every leaf of the minimal marked tree is marked, so the only possible leaves are A_k and P_1. They are therefore the two leaves and the reduced tree is their path. Let T_* be the singleton triangle nearest P_1. Cut an actual bridge immediately beyond the terminal subtree containing T_* and P_1. Lemma 2.1 gives

$$TP + (\text{connected pentacyclic cactus}).$$

The first territory has surplus $> 1 - \delta > 0$ and the second is strictly positive. This includes the all-singleton path $P - T - T - T - T - T - P$. The construction never splits A_k, so its incidence and the entry by which the path reaches it are immaterial. □

Lemma 6.2 (The $T^5P \mid P$ degree dichotomy)**.** *Every cactus with cluster partition $T^5P_0 \mid P_1$ has positive surplus.*

Proof. Let I be the incidence tree of the cluster T^5P_0 and put $d = \deg_I(P_0)$. Project the external connector entry to the first point where it reaches the cyclic hull; an entry through a hanging branch follows the root of that branch.

Suppose first that $d \geq 2$. The components of $I - P_0$ contain respectively $r_1, \ldots, r_d \geq 1$ triangles, with sum five. Split P_0 into one proper consecutive interval per component. Adjoin the entire external connector and P_1 to the interval owning its entry. If that branch contains r triangles, its retained territory has type $T^r P_1$: for $r = 1$ it is a strict TP packet, for $r = 2$ it is a nonnegative tricyclic cactus, and for $r \geq 3$ it is strictly positive by (5). Every other interval territory is an A_{r_j} packet and is strict. Since $d \geq 2$, at least one such other territory exists, so the sum is strictly positive even when the entry territory has only the nonnegative tricyclic bound. If the entry is a private vertex of P_0, assign it and the connector to either adjacent interval; at a shared cut or through a triangular branch the incidence component determines its unique owner. This also covers the saturated case $d = 5$.

Now suppose $d = 1$. Choose a private vertex v_0 of P_0 away from the connector entry and a private vertex v_1 of P_1 away from its entry. Such choices exist: P_0 has four vertices private with respect to cyclic blocks, and P_1 has no shared cyclic cut. Open both pentagons at these vertices. The two opened territories F_0, F_1 are disjoint nonempty trees with surplus -1 each. All remaining vertices form a connected induced territory H: the path $P_0 - v_0$ remains attached at its unique cyclic cut, the path $P_1 - v_1$ remains joined through the connector, and the choices away from the entries preserve the connector route. The only cycles of H are the five triangles. Deleting the incidence-leaf P_0 leaves them in one shared-cut cluster, so Lemma 3.1 gives

$$\sigma(G) \geq \sigma(H) + \sigma(F_0) + \sigma(F_1) > 2 - 1 - 1 = 0.$$

Thus both possible degrees, every cyclic order, and every connector entry are covered. □

Proposition 6.3. *If the cycles are T^5PP and the shared-cut graph is disconnected, then $s^+(G) > |V(G)|$.*

Proof. The colored recursion gives all 46 proper partitions. The packet ledger settles the displayed 41 direct rows. Lemma 6.1 settles the first four remaining rows and Lemma 6.2 settles the fifth. Hence every colored partition, reduced-tree topology, connector entry, and internal cluster incidence is covered. □

7 One fully shared T^6Q cluster

Proposition 7.1 (Leaf Q or internal Q split)**.** *If all seven T^6Q cycles form one shared-cut cluster, then $s^+(G) > |V(G)|$.*

Proof. Let I be the cycle–cut incidence tree and inspect the designated node Q. If $\deg_I(Q) = 1$, open a vertex of Q private with respect to cyclic blocks. Such a vertex exists because $q \geq 3$. The opened territory is a nonempty tree of surplus -1. Deleting the incidence-leaf Q leaves the six triangle nodes connected in one shared-cut cluster, so the complementary induced territory is an A_6 packet. Therefore

$$\sigma(G) \geq \sigma(A_6) - 1 > 1 - 1 = 0.$$

If $d = \deg_I(Q) \geq 2$, split Q into one proper consecutive interval per component of $I - Q$. Each component contains at least one triangle: the cut neighbor of Q has degree at least two, and no other nontriangle exists. If their triangle counts are $r_1, \ldots, r_d$, then $r_j \geq 1$ and $\sum_j r_j = 6$. The exact induced territories are $A_{r_1}, \ldots, A_{r_d}$, with the path intervals of Q and all hanging trees absorbed as attachments. By Lemma 3.1,

$$\sigma(G) \geq \sum_j \sigma(A_{r_j}) > \sum_j b_{r_j} \geq 0.$$

Strictness follows from the strict estimate for every branch. This case includes a saturated Q hub and all dispersed branch patterns. The seven-cycle bouquet lies in the first case: opening Q leaves the six-triangle common-cut packet and closes it with one tree cost. □

For comparison and as an independent finite certificate, let c be the number of shared cyclic cuts. The incidence tree has $c + 6$ edges and

$$\sum_{x \text{ cut}} (\deg_I(x) - 1) = 6, \qquad 1 \leq c \leq 6.$$

Quotienting by permutations of the six triangles and cut nodes, while keeping Q distinguished even when $q = 3$, gives the corrected census

Q capacity	$c=1$	$c=2$	$c=3$	$c=4$	$c=5$	$c=6$	total
$q=3$	1	8	33	71	74	29	216
$q=4$	1	8	33	73	77	32	224
$q=5$	1	8	33	73	78	33	226
$q=6$	1	8	33	73	78	34	227
$q\geq 7$	1	8	33	73	78	34	227

An exact conservative packet test finds a safe ordinary one-cycle split in every nonbouquet class, respectively $215, 223, 225, 226, 226$ classes. The sole ordinary-split exception in every regime is the seven-cycle bouquet already handled in Proposition 7.1. These counts are a certificate of the structural proof, not a premise of it.

8 One fully shared T^5PP cluster

Proposition 8.1 (Two pentagonal leaves or an internal pentagon)**.** *If all seven T^5PP cycles form one shared-cut cluster, then $s^+(G) > |V(G)|$.*

Proof. Let I be the incidence tree and first suppose both pentagon nodes are leaves. Open one private vertex on each pentagon. The opened territories are disjoint nonempty trees of surplus -1. Successively deleting the two leaf cycle nodes from the tree leaves the five triangle nodes connected in one shared-cut incidence tree. The complementary territory is therefore an A_5 packet, and

$$\sigma(G) \geq \sigma(A_5) - 2 > 2 - 2 = 0.$$

The two pentagons are distinct blocks, so their opened territories are disjoint; each private choice can include all tree branches rooted there.

Otherwise an internal pentagon P_0 exists. Put $2 \leq d = \deg_I(P_0) \leq 5$ and split P_0 into one proper consecutive interval per component of $I - P_0$. There is a unique branch B containing the other pentagon P_1; let a be the number of triangles in B.

If $a \geq 1$, then $a \leq 4$, because at least one other incidence component contains a cycle and every other cycle is triangular. The B territory has type T^aP: it is strictly positive for $a = 1$ by (3), nonnegative for $a = 2$ by (4), and strictly positive for $a = 3, 4$ by (5). At least one other branch is an all-triangle packet and is strict by Lemma 3.1. The total is positive.

If $a = 0$, then B is a singleton pentagon and contributes at least $-\delta$. The five triangles occupy at most $d-1 \leq 4$ other nonempty branches. Some branch therefore contains at least two triangles; its shared-triangle margin is > 1 by Lemma 3.1. Every remaining triangular branch is strict, and the total is $> 1-\delta > 0$. This includes a saturated pentagon hub: if the other pentagon is a singleton petal, the five triangles in at most four other branches force the required TT-or-larger packet. The two cases exhaust the incidence tree and every interval has one owner by Lemma 2.2. □

The independent exact census provides a second completeness check. Quotient by $S_5 \times S_2 \times S_c$ and test every cycle deletion against the retained incidence hypotheses of the packet ledger. The exact result is

c	1	2	3	4	5	6	total
colored incidence trees	1	12	68	177	211	91	560
SAFE ordinary split	0	11	67	177	211	91	557

The three canonical ordinary-split exceptions are exactly

$$((0,7),(1,7),(2,7),(3,7),(4,7),(5,7),(6,7)), \tag{14}$$
$$((0,7),(1,7),(2,7),(3,7),(4,7),(5,7),(0,8),(6,8)), \tag{15}$$
$$((0,7),(1,7),(2,7),(3,7),(4,7),(0,8),(5,8),(1,9),(6,9)), \tag{16}$$

where $0,\ldots,4 = T$, $5,6 = P$, and cut nodes begin at 7. They are the seven-cycle bouquet, a common-cut T^5P core with a TP tail, and a five-triangle common-cut core with two separate pentagon tails. In all three, both pentagons are incidence leaves. Thus the first case of Proposition 8.1 opens them and leaves one concentrated A_5 packet. In particular, the exact $560 = 557+3$ certificate has no unresolved saturated hub. The structural proposition proves all classes without relying on the census.

9 Main theorem

Theorem 9.1. *Every connected heptacyclic cactus G on n vertices satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. The exact DNN estimate (4.1) is strict unless the cycle multiset is T^6Q or T^5PP. For T^6Q, a disconnected shared-cut graph is handled by Proposition 5.1, while one shared-cut cluster is handled by Proposition 7.1. For T^5PP, the corresponding cases are Propositions 6.3 and 8.1. These alternatives exhaust every cycle multiset and every shared-cut incidence.

Lemmas 2.1 and 2.2 assign every actual connector segment, Steiner branch, entry vertex, shared cyclic cut, and hanging tree to exactly one territory. Private openings are paid only by the explicit shared-triangle margins of Lemma 3.1. Every spectral comparison is applied to a vertex partition into induced subgraphs; no edge monotonicity, tree relocation, bounded connector length, or special attachment position is assumed. □

Since a connected heptacyclic graph has $|E(G)| = n+6$, this proves the strict conclusion of the Akbari–Kumar–Mohar–Pragada–Zhang conjecture [2, Conjecture 1.2] for every heptacyclic cactus.

A Exact reproduction and proof objects

From the repository root run, using only the Python standard library,

```
python research/heptacyclic-t5pp-disconnected-partition-audit.py
python research/heptacyclic-t6q-incidence-census.py
python research/heptacyclic-tttttpp-incidence-census.py
```

All numerical proof comparisons in these scripts use `fractions.Fraction`; no floating-point comparison is a premise. The partition script asserts

```
colored partitions including one cluster: 47
proper colored partitions: 46
direct packet rows: 41
topology/entry rows: 5
```

and asserts exactly the five rows in (13).

The T^6Q script asserts, by cut count $c = 1,\ldots,6$,

```
q=3:  1, 8, 33, 71, 74, 29 = 216
q=4:  1, 8, 33, 73, 77, 32 = 224
q=5:  1, 8, 33, 73, 78, 33 = 226
q=6:  1, 8, 33, 73, 78, 34 = 227
q>=7: 1, 8, 33, 73, 78, 34 = 227
```

and safe-split totals $215, 223, 225, 226, 226$, with the bouquet as the unique exception in each regime. Generation is by exhaustive leaf extension of the rank-six T^5Q incidence classes, followed by an independent center-rooted color-preserving tree code.

The T^5PP script asserts

```
colored trees by cut count:       1, 12, 68, 177, 211, 91 = 560
SAFE-resolved trees by cut count: 0, 11, 67, 177, 211, 91 = 557
unresolved structural classes: bouquet, T^5P core with TP tail,
                               T^5 core with two P tails
```

It checks all $7 \cdot 560 = 3920$ cycle choices and verifies the retained shared-cut hypotheses before using concentrated packet bounds. Its rational proxies include $P > -1/4$, $TP > 3/4$, shared $TTP > 7/4$, and shared $TPP > 3/2$. These follow exactly from $\delta < 1/4$ and $6-2\sqrt{5} > 3/2$, certified by squaring positive rational bounds.

The scripts enumerate finite colored incidence objects or colored cluster partitions. They do not infer inducedness from abstract trees. Induced realization, cyclic intervals, connector ownership, arbitrary attached trees, and the structural repairs are proved in the body of the manuscript. The current source proof objects are

```
research/heptacyclic-dnn-residuals-2026-07-26.md
research/heptacyclic-t6q-disconnected-2026-07-26.md
research/heptacyclic-t5pp-disconnected-complete-audit-2026-07-26.md
research/heptacyclic-residual-sacrifice-splitting-lemmas-2026-07-26.md
research/heptacyclic-cactus-complete-synthesis-2026-07-26.md
research/heptacyclic-t5pp-disconnected-partition-audit.py
research/heptacyclic-t6q-incidence-census.py
research/heptacyclic-tttttpp-incidence-census.py
```

The corrected totals displayed here are generated by these current scripts; no earlier pre-fix census count is used.

References

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[4] Companion manuscript, *Positive Square Energy of Every Tricyclic Cactus*, 2026; repository path `all-tricyclic-cacti/paper.tex`.

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